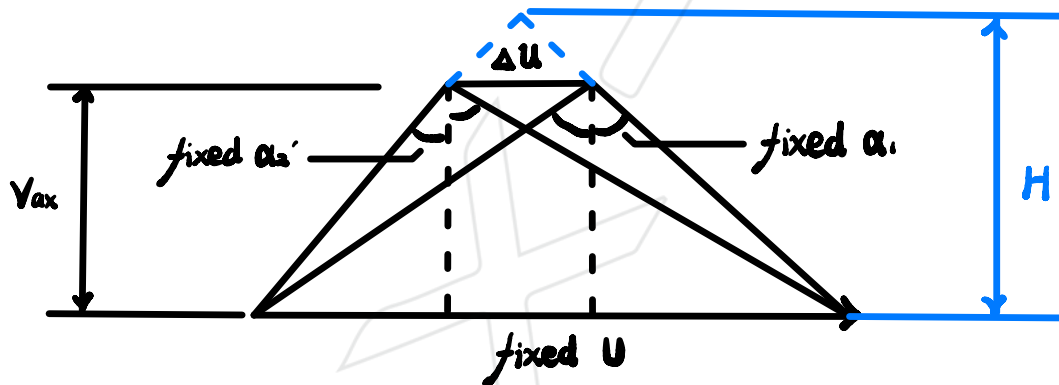


AERO50009 Propulsion and Turbomachinery
Tutorial Sheet 4

Consider a compressor consisting of a row of inlet guide vanes, followed by a single rotor, and an outlet stator. The inlet guide vanes ensure that the angle of the velocity upstream of the rotor is fixed and equal to α_1 . The shape of the rotor blades ensures that the velocity angle downstream of the rotor in the rotor frame of reference is fixed and equal to α_2' . The compressor is driven by an electric motor with a fixed rpm, so that the velocity of the rotor is fixed and equal to U . The compressor is used as a part of a hydraulic drive: the compressor outlet is connected via a pipe to a hydraulic motor. Assume that the fluid is incompressible. What should be the axial velocity for passing the maximum power to the hydraulic motor?



$$\Delta u = \dot{m}_b V_{ax} (\tan \alpha_1' - \tan \alpha_2') = \dot{m}_b V_{ax} (\tan \alpha_2 - \tan \alpha_1)$$

$$= \dot{m}_b [U - V_{ax} (\tan \alpha_1' + \tan \alpha_2')] \quad \text{with } \dot{m}_b = \rho V_{ax} A$$

$$\dot{W} = \dot{m} C_p (T_{03} - T_{01}) = \dot{m} U [U - V_{ax} (\tan \alpha_1' + \tan \alpha_2')] \quad \text{with } \dot{m} = \rho V_{ax} A$$

$$\dot{W} = \dot{m} U \Delta u = \dot{m} U (H - V_{ax}) (\tan \alpha_1 + \tan \alpha_2')$$

$$= \rho V_{ax} A (H - V_{ax}) (\tan \alpha_1 + \tan \alpha_2')$$

→ parabolic curve for $\dot{W}$ versus V_{ax}